C.W

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⊞ Show that standard deviation.

$S.D(\sigma) = \sqrt{\frac{1}{12}(n^2-1)}$, where n is natural number.

Proof:

Consider, first n natural numbers as x.

According by definition of standard deviation,

$$S.D(\sigma) = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Sum of first n natural number,

$$\sum x = 1 + 2 + 3 + \dots + n$$

$$= \frac{n(n+1)}{2}$$

Sum of the square of a n natural number,

$$\sum x^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2$$

$$= \frac{n(n+1)(2n+1)}{6}$$

$$\therefore S.D(\sigma) = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Now,

$$\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

$$= \frac{n(n+1)(2n+1)}{6n} - \left\{\frac{n(n+1)}{2n}\right\}^2$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= \frac{(n+1)}{2}\left\{\frac{(2n+1)}{3} - \frac{(n+1)}{2}\right\}$$

$$= \frac{(n+1)}{2}\left\{\frac{2(2n+1) - 3(n+1)}{6}\right\}$$

$$= \frac{(n+1)}{2}\left\{\frac{4n+2-3n-3}{6}\right\}$$

$$= \frac{(n+1)}{2} \cdot \frac{(n-1)}{6} = \frac{(n+1)(n-1)}{12} = \frac{1}{12}(n^2-1)$$

So, S.D $(\sigma) = \sqrt{\frac{1}{12}(n^2-1)}$

(Showed)

(*) Find IQR for the given data:

Day	0-4	4-8	8-12	12-16	16-20	$n=50$
absent	5	12	23	8	2	
c.f.	5	17	40	48	50	

Solution:

$\therefore Q_1 = L + \frac{\frac{n}{4} - pcf}{f} \times c$

$= 4 + \frac{12.5 - 5}{12} \times 4$

$= 4 + \frac{7.5}{12} \times 4$

$= 4 + 2.5$

$= 6.5$

Here,
$\frac{n}{4} = \frac{50}{4} = 12.5$
$L = 4$
$pcf = 5$
$f = 12$
$c = 4$

$\therefore Q_3 = L + \frac{\frac{3n}{4} - pcf}{f} \times c$

$= 8 + \frac{37.5 - 17}{23} \times 4$

$= 8 + \frac{20.5}{23} \times 4$

$= 8 + 3.5 = 11.5$

Here,
$\frac{3n}{4} = \frac{3 \cdot 50}{4} = 37.5$
$L = 8$
$pcf = 17$
$f = 23$
$c = 4$

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$\therefore IQR = Q_3 - Q_1$

$= 11.5 - 6.5 = 5.1$

আগামীকাল পরিক্ষার জন্য Revise দিবে আমরা শুরু থেকে শেষ পর্যন্ত।